

Self-Organizing Map for Iris Dataset Classification



Figure 1: Self-organizing map applied to the Iris dataset showing winning neuron positions in the 40×40 output array. **Left panel:** distribution with randomly initialized weights. **Right panel:** distribution after 10 epochs of training using Kohonen’s learning rule with Gaussian neighborhood function. Colors indicate species: Iris-setosa (red), Iris-versicolor (blue), Iris-virginica (green). Both axes show neuron positions in the grid.

Implementation. I implemented a self-organizing map with a 40×40 neuron grid to cluster the Iris dataset. The network was trained for 10 epochs using the learning rule from Eq. (10.17), with Gaussian neighborhood function (10.18). Training parameters were: $\eta_0 = 0.1$, $d_\eta = 0.01$, $\sigma_0 = 10$, $d_\sigma = 0.05$. Input data were normalized by dividing by the maximum value. Weights were initialized from uniform distribution $[0, 1]$.

The left panel of Figure 1 shows the random distribution of winning neurons before training, with all three classes mixed across the grid. After training, the SOM successfully organized the samples into three distinct clusters. Setosa (red) formed a well-separated cluster in the upper-right region, while Versicolor (blue) occupied the center region, and Virginica (green) the lower-left area, with some overlap between the latter two species.

Discussion. The clustering demonstrates effective topographic mapping, with the SOM reducing dimensionality from 4D to 2D while preserving class separability. The clear separation of Iris-setosa reflects its uniqueness in the features, while the partial overlap between Iris-versicolor and Iris-virginica is consistent with their greater similarity.

Potential sources of inaccuracy include: 10 epochs is very limited, it could cause incomplete convergence at cluster boundaries; the exponential decay schedule, which reduces the neighborhood influence relatively quickly and may cause premature convergence; and the finite grid resolution, which constrains the spatial precision of the mapping.

Conclusions. The three clusters are distinguishable in Figure 1, confirming successful unsupervised learning of the class structure. The results align with expected behavior for the Iris dataset, where setosa is known to be easily separable while versicolor and virginica share more similar characteristics.

Part of the code structure was organized with the assistance of OpenAI’s ChatGPT (GPT-5), also used for troubleshooting implementation errors.